



CAMARENA Francesca

Robotics | Software Engineer | Mechatronics Engineer

Experience

02/19 - Now **Senior Software Engineer for Autonomous Driving**

BMW Group, Munich, Germany

I work on Obstacle Avoidance feature as part of a team. In addition to that, I am Deputy Facilitator of Software Quality community, one of a handfull of people defining standards for C++ within the company, and I wrote the library for error handling mechanism used throughout the BMW Group codebase.

07/17 - 12/17 **Robotics Software Engineer Intern**

Nuro, Mountain View, USA

I worked as part of a small perception team developing the first ever autonomous delivery bot. My work focused on LiDAR scene segmentation, sensor fusion, and sensor calibration. I was fully responsible for all software for cameras and LiDAR calibration.

05/14 - 02/19 **Research Fellow**

University of Bonn, Bonn, Germany

I was the first student at the lab and have set up all the lab infrastructure including a full CI setup. Scientifically I have worked on various topics such as mobile robot navigation, range-based scene interpretation, and mapping. I published at top-tier international conferences such as ICRA and IROS and my code is available as open source.

01/12 - 05/14 **Research Assistant**

AIS lab, University of Freiburg, Freiburg, Germany

As a research assistant in the Autonomous Intelligent Systems lab at the University of Freiburg, led by Prof. Dr. Wolfram Burgard, I dealt with Kinect RGBD sensors mounted onto various platforms. I have implemented traversability analysis for a mobile robot as part of the ROVINA project. The developments in this project led to a publication at ECMR already during my masters.

12/11 - 03/12 **Tutor, Image Processing**

University of Freiburg, Freiburg, Germany

During my first semester I was tutoring the Image Processing course.

03/10 - 10/11 **Junior Software Engineer**

Timecode LLC, Kyiv, Ukraine

Developed casual games for Android and a Web store for ASUS Xtion.

Selected Projects

Address

Paris, France

On the Web

GitHub: :cuycorp
LinkedIn: :Francesca
Camarena

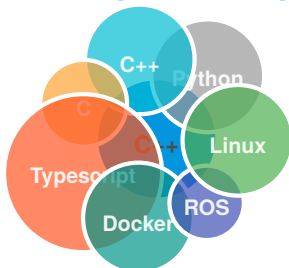
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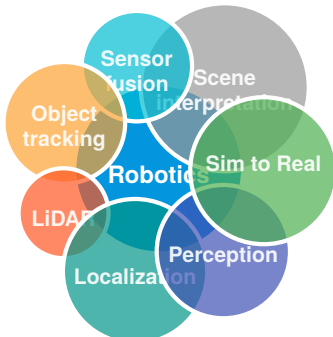
Mail

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Programming



Fields of Interest



Languages

English [bilingual] ★★★★★
French [proficient] ★★★★★
Spanish [native] ★★★★★

2026 - Present | **Modern Robot Learning**

[AI Lab - 42 AI](#)

A modular robotics research platform for learning and experimentally evaluating modern approaches to robot perception, manipulation, imitation learning, reinforcement learning, and embodied foundation models.

<https://github.com/cuycorp/Modern-Robot-Learning>

2026 - Present | **Duck - a thought tracker**

[Personal project](#)

A local-first focus companion, desktop overlay application, that lets users quickly externalize distracting thoughts and return to their task.

<https://github.com/cuycorp/Duck>

2026 - Present | **Tokenscope**

[Ecole 42](#)

TokenScope is a collaborative platform that helps engineering teams understand where their LLM spending comes from by collecting usage traces, calculating estimated cost, visualizing live analytics, and generating grounded optimization recommendations.

<https://github.com/lynyam/tokenscope>

2025 | **Minishell**

[Ecole 42](#)

Developing an autonomous robot for underground exploration I was responsible for traversability analysis, navigation, exploration and some of the mapping tasks. The project received excellent reviews from the EU commission.

<https://github.com/cuycorp/minishell>

Video Lectures on C++ on YouTube

I have designed and released a course on Modern C++ for Image Processing. It has more than 45000 views to date and gains popularity in the robotics circles. It is available on **YouTube** and on the website of the IPB lab:
<http://www.ipb.uni-bonn.de/teaching/modern-cpp/>.

First-author Publications

A general framework for flexible multi-cue photometric point cloud registration	ICRA 2018
Analyzing the quality of matched 3d point clouds of objects	IROS 2017
Efficient Online Segmentation for Sparse 3D Laser Scans	PFG 2017
Fast range image-based segmentation of sparse 3d laser scans for online operation	IROS 2016
Robust homing for autonomous robots	ICRA 2016
Where to Park? Minimizing the Expected Time to Find a Parking Space	ICRA 2015
Efficient Traversability Analysis for Mobile Robots using the Kinect Sensor	ECMR 2013

Honors & Awards

2013 MINT Excellence Network Member	MLP Stiftung
I was chosen as one of 300 best applicants across Germany to the MINT Excellence Network. The candidates were chosen from the students who work in the fields of Maths, Computer Science, Natural Sciences and Tech across Germany.	
2018 Best PhD Thesis of the Faculty Award	University of Bonn
My PhD thesis "Robot mapping and navigation in real-world environments" has been awarded the title of the best out of all written at my faculty in 2018.	

Education

2011 - 2014 MSc. Computer Software Engineering	Ecole 42, France
A project-based and peer-to-peer computer science program ranked the world's 3rd most innovative university in the 2025 and 2026 WURI rankings. Training focused on system-level programming, C, Linux, Git, and shell development, with extensive team projects in an autonomous, collaborative learning environment.	
2017- 2022 BSc. Mechatronics Engineering	Pontifical Catholic University of Peru, Lima, Peru
The Pontifical Catholic University of Peru (PUCP) is ranked No. 1 in Peru and No. 10 in Latin America. My studies focused on mechanical and electronic systems and the design of integrated hardware–software automation systems. Thesis: design of an EEG-controlled omnidirectional wheelchair..	
2015 - 2016 Bilingual International Baccalaureate Diploma	Newton College, Lima, Peru
Top bilingual Peruvian-British school, focused on developing inquisitive, outspoken and well-rounded individuals that have intercultural understanding.	

September 28, 2026

CAMARENA PUCUHUANCA, Monica Francesca